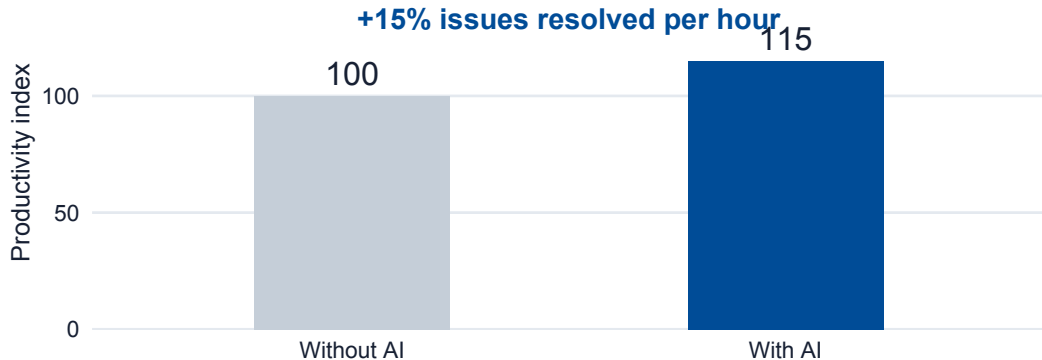


AI and Labor Productivity

5,172 customer-support agents · one firm · baseline normalized to 100



Source: Brynjolfsson, Li, and Raymond (2025), Quarterly Journal of Economics. Bars translate the estimated effect into an index.

AI and Economic Growth

- ▶ **Adoption takes investment.** Firms need equipment, training, and reliable ways to check the output.
- ▶ **Tasks and jobs change.** AI can help workers and replace some tasks. Higher output can coexist with job losses in particular occupations.
- ▶ **Benefits and costs are shared unevenly.** Workers, owners, and consumers may gain differently. Computing also requires electricity and infrastructure.

Source: Acemoglu (2024), The Simple Macroeconomics of AI; IEA (2025), Energy and AI.

The Aggregate Production Function

The production function links output to inputs and technology. In output-per-worker form:

$$y = A f(k, h)$$

1. k : **physical capital per worker**. AI servers and computing equipment.
2. h : **human capital per worker**. Skills to use AI and check its work.
3. A : **technology**. Better methods that produce more with the same inputs.

y is real output per worker. To study one channel, hold the others fixed.

Measuring the Sources of Growth

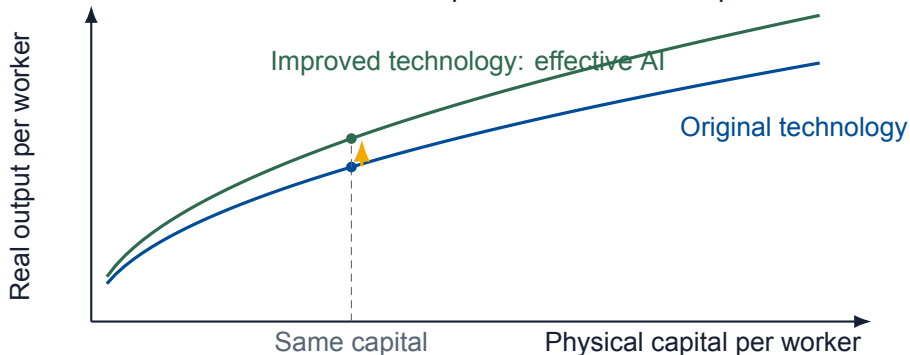
- ▶ **Physical capital (k):** Estimate equipment and buildings from past investment, adjusting for prices and depreciation. Divide by the number of workers.
- ▶ **Human capital (h):** Use years of schooling and estimated returns to education to approximate workers' skills. Schooling is an imperfect measure.
- ▶ **Total factor productivity (A):** Estimate the growth in output per worker left after subtracting the **weighted contributions** of capital and skills per worker.

This **residual** includes technological progress, but also changes in efficiency and measurement errors.

Source: Penn World Table methodology, Feenstra, Inklaar, and Timmer (2015); BLS, Handbook of Methods.

Technological Progress and Output

Effective AI methods can raise output with the same capital and skills.



Schematic production functions; human capital held fixed